

Challenge Write-Up Template

Challenge Information

Student Name	Daniella Marie A. Leonud
Section / Class	BSIT-3G
Challenge Name	Undo
Category	General Skills
Points / Difficulty	Easy
Date Solved	August 10, 2026
Flag Format Given	picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_8deed4b6}

1. Challenge Description

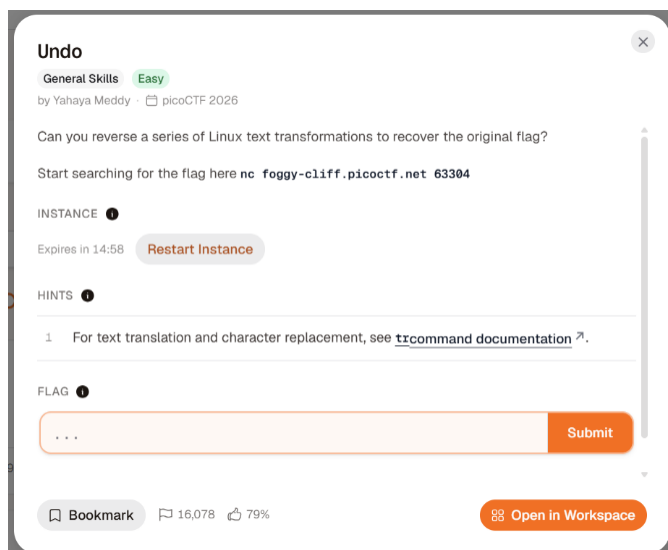
Restate the challenge in your own words: what were you given, and what were you asked to find or do? Do not simply copy-paste the original prompt.

I was given a challenge that showed an encoded string and told me it had undergone several text changes. My goal was to answer each step with the right Linux command to reverse those changes one by one and get the original flag back.

2. Initial Analysis / Reconnaissance

What did you check first? What did file, strings, a webpage view-source, or another first look tell you? What category did you decide this was, and why?

First, I opened and read the challenge and the hint provided. The hint contained a link to a Linux manual page, which was helpful because I could use it as a reference while solving the challenge. After that, I launched the instance. The server gave me the following netcat command **nc foggy-cliff.picoctf.net 63304** I copied the command and entered it into the terminal to connect to the challenge server. Once I connected, the server gave me a series of instructions. Each instruction described a transformation that had been done to the text. My job was to reverse each one using the appropriate Linux command.



3. Tools and Commands Used

List every tool/command you used, in the order you used them, and what each one told you.

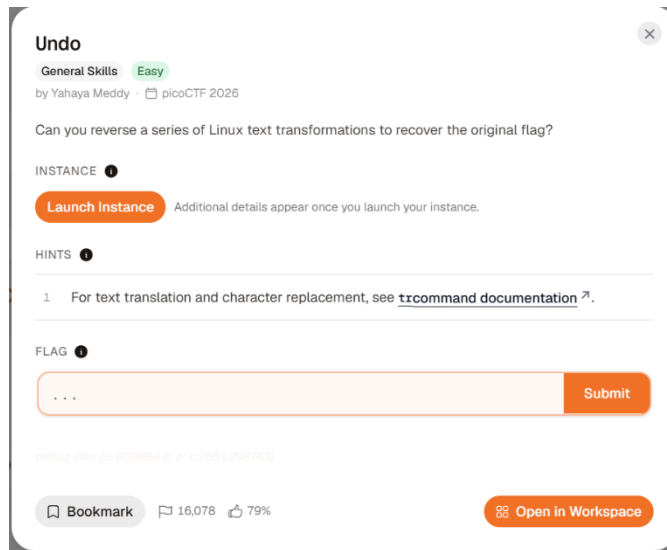
Tool / Command	What it was used for	Result / Output
nc foggy-cliff.picoctf.net 63304	Connected to the challenge server.	Opened the interactive challenge prompt.
base64 -d	Decoded the initial Base64 text string.	Reversed Step 1; gave the text for Step 2.
rev	Reversed the order of the characters back to normal.	Reversed Step 2; gave the text for Step 3.
tr '-' '_'	Changed all dashes (-) back into underscores (_).	Reversed Step 3; gave the text for Step 4.
tr '()' '{}'	Changed parentheses () back into curly braces {}.	Reversed Step 4; gave the text for Step 5
tr 'a-zA-Z' 'n-za-mN-ZA-M'	Decoded the ROT13 letter shift.	Reversed Step 5; revealed the final flag.

4. Solution Walkthrough

Explain your reasoning step by step — not just what you typed, but why you tried it and what you concluded from the result. Include screenshots, code snippets, or command output where useful. A reader who has not seen the challenge should be able to follow your logic and arrive at the same answer.

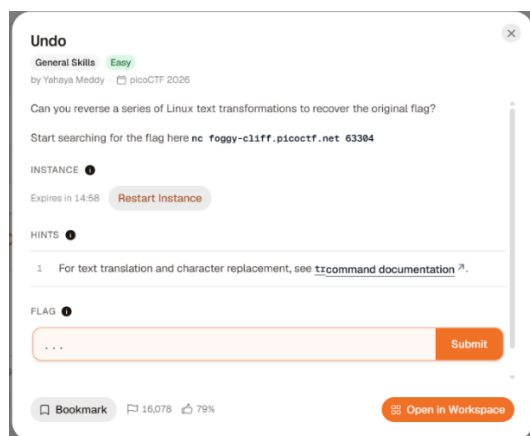
Step 1

First, I opened and read the challenge and the hint provided. The hint directed me to a Linux manual page. I opened the link and saw that it contained information about Linux commands. This was helpful because the challenge required me to use Linux text-processing commands to reverse the transformations.



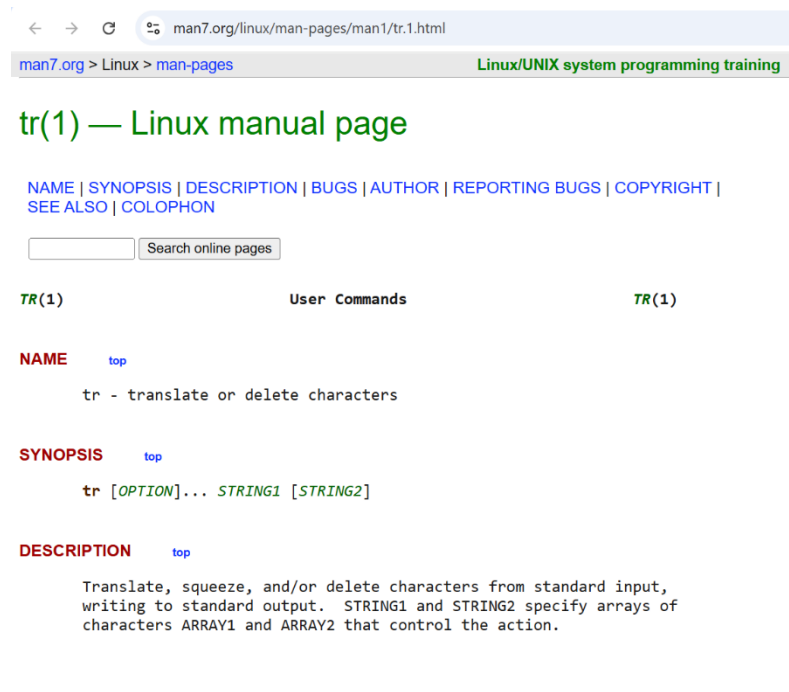
Step 2

After launching the instance, the server gave me the command **nc foggy-cliff.picoctf.net 63304**. I copied the command and pasted it into the terminal. The Netcat command connected me to the remote challenge server and opened an interactive prompt. From there, the server started giving me the different transformations that I needed to undo.



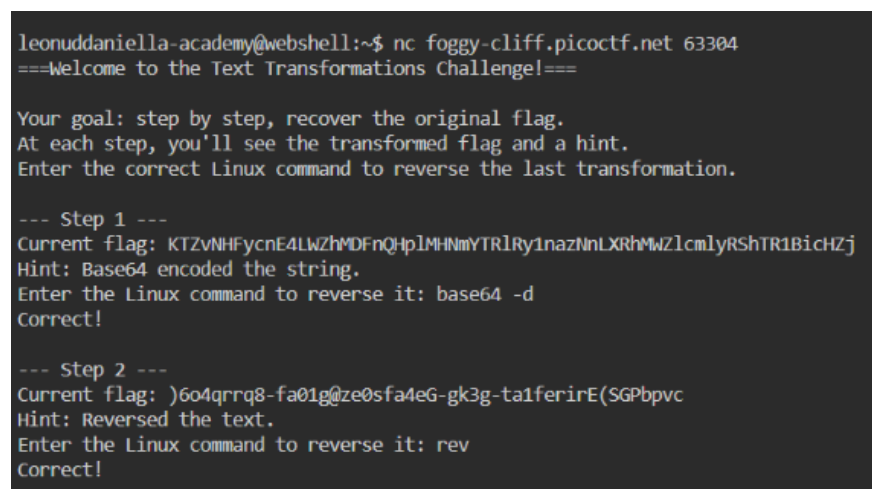
Step 3

This is what the link looks like when I opened it. It is the linux manual page, and this is very helpful to me when I answer this challenge.



Step 4

I started by pasting the **nc foggy-cliff.picocft.net 63304** and was welcomed to the challenge. I read the hint given and the server asked me to reverse Base64 encoding. I entered **base64 -d** to decode it. Next, the server said the text had been reversed. I used the **rev** command to flip the characters back to their normal order.



Step 5

Next, replace the underscore with dashes and to swap them back, I ran `tr '-' '_'` to replace every - with an _. `tr` means translate. So, the same as the step 3 the hint stated that curly braces were swapped with parentheses. I ran `tr '()' '{}'` to replace (and) with { and }.

```
--- Step 3 ---
Current flag: cvpbPGS(Ereref1at-g3kg-Ge4afs0ez@g10af-8qrrq4o6)
Hint: Replaced underscores with dashes.
Enter the Linux command to reverse it: tr '-' '_'
Correct!

--- Step 4 ---
Current flag: cvpbPGS(Ereref1at_g3kg_Ge4afs0ez@g10af_8qrrq4o6)
Hint: Replaced curly braces with parentheses.
Enter the Linux command to reverse it: tr '()' '{}'
```

Step 6

Now, for step 5. I first ran `tr 'A-Za-z' 'N-ZA-Mn-za-m'` but it was incorrect, by reading the hint and copying it, then it was finally as success. The correct command is `tr 'a-zA-Z' 'n-za-mN-ZA-M'`

```
--- Step 5 ---
Current flag: cvpbPGS{Ereref1at_g3kg_Ge4afs0ez@g10af_8qrrq4o6}
Hint: Applied ROT13 to letters.
Enter the Linux command to reverse it: tr 'A-Za-Za-z' 'N-ZA-Mn-za-m'
Incorrect. Try again.
Output: [Error] tr: range-endpoints of 'a-Z' are in reverse collating sequence order
Hint: Try reversing: tr 'a-zA-Z' 'n-za-mN-ZA-M'

Enter the Linux command to reverse it: tr 'a-zA-Z' 'n-za-mN-ZA-M'
Correct!

Congratulations! You've recovered the original flag:
>>> picoCTF{ReversIng_t3xt_Tr4nsf0rm@t10ns_8deed4b6}
```

Step 7

I copied the flag given by the server and pasted it into the submission box. The challenge accepted the flag, confirming that I had completed the Undo challenge

Undo

General Skills Easy
by Yahaya Meddy · picoCTF 2026

Launch Instance

Additional details appear once you launch your instance.

HINTS 1

1 Reveal hint

FLAG 1

picoCTF{ReversIng_t3xt_Tr4nsf0rm@t10ns_8deed4b6}

Submit

Correct flag!

5. Flag

```
picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_8deed4b6}
```

6. Key Concept / What I Learned

In 2-4 sentences, name the underlying security concept or vulnerability class this challenge taught (e.g. "weak RSA exponent," "reflected XSS," "stack buffer overflow"). This is the part of the write-up most worth remembering after the flag is submitted.

In this challenge, I learned how to use common Linux text-processing commands to undo basic data transformations. I learned how tools like base64, rev, and tr work together to clean up or decode encoded strings directly from the terminal. Having the manual actually makes this challenge easier because it guides me throughout it. Even though I ran an incorrect command, it gave me a hint on what I should enter to get it right.

7. References

List any external resource you consulted beyond course materials (a tool's documentation, a technique explainer, a similar public write-up). It is normal and expected to use references — just cite them.

Linux tr Manual (<https://man7.org/linux/man-pages/man1/tr.1.html>)

Linux rev Manual (<https://man7.org/linux/man-pages/man1/rev.1.html>)

GNU Coreutils — Base64 (https://www.gnu.org/software/coreutils/manual/html_node/base64-invocation.html)
